



Meridian Junior College
JC2 Preliminary
Examinations

Name:

Class:

Index No:

BIOLOGY

9747/02

Paper 2 Core Paper

08 Sep 2008

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and register number on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination,

1. Write your name, index number and civics group on all sheets of writing papers.
2. Fasten all your writing papers together.
3. Hand in Section A and Section B separately.
4. Write down the question number for the question answered for Section B.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	
Total	

Section A

Answer **all** questions in this section.

1. Fig 1.1 is an electron micrograph of a plasma cell from bone marrow.

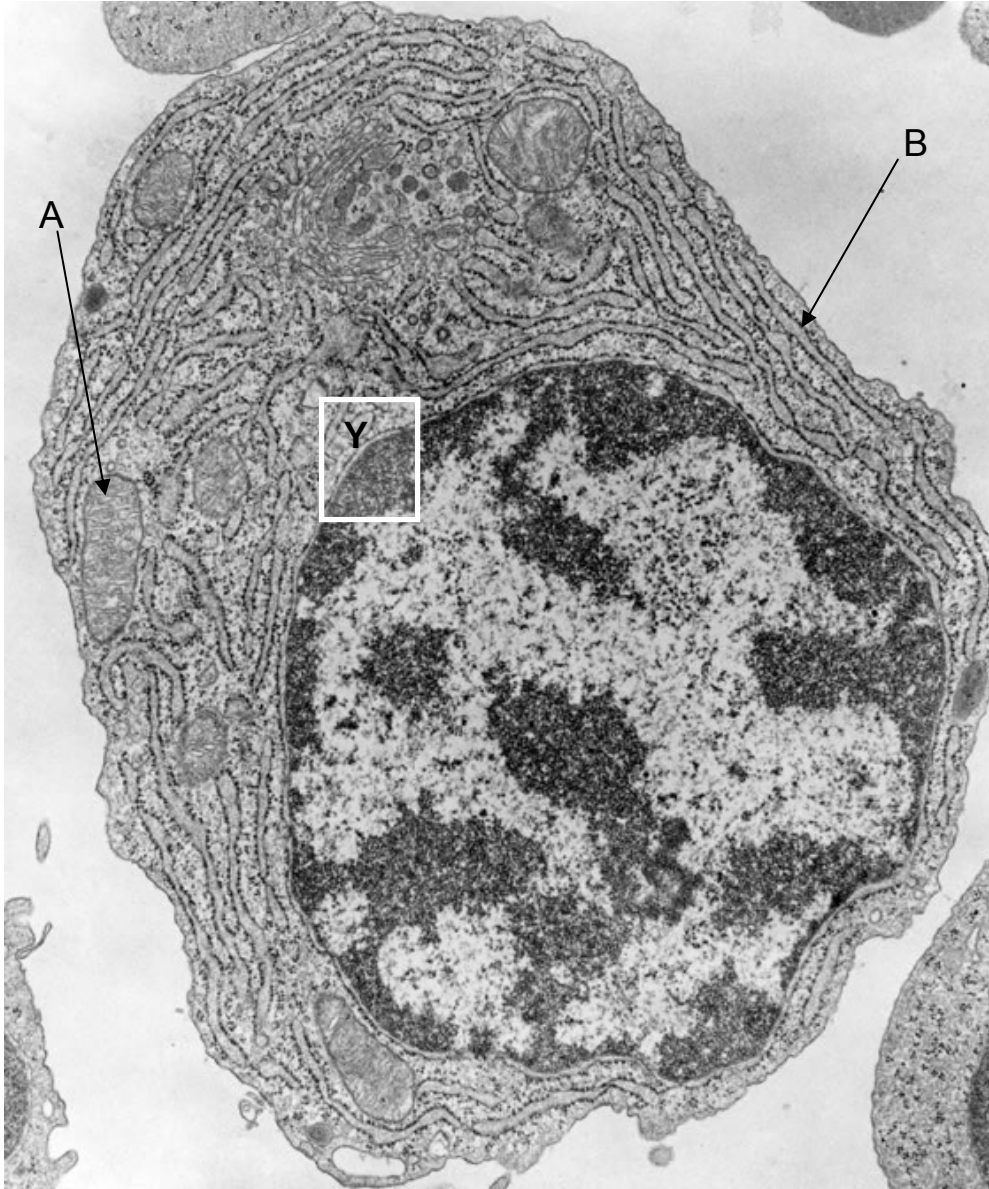


Fig. 1.1

(a) Identify organelles A and B.

[2]

(b) With reference to Fig. 1.1, briefly describe the structural features of organelle B. [2]

Fig. 1.2 shows a diagrammatic representation of Region Y in Fig. 1.1.

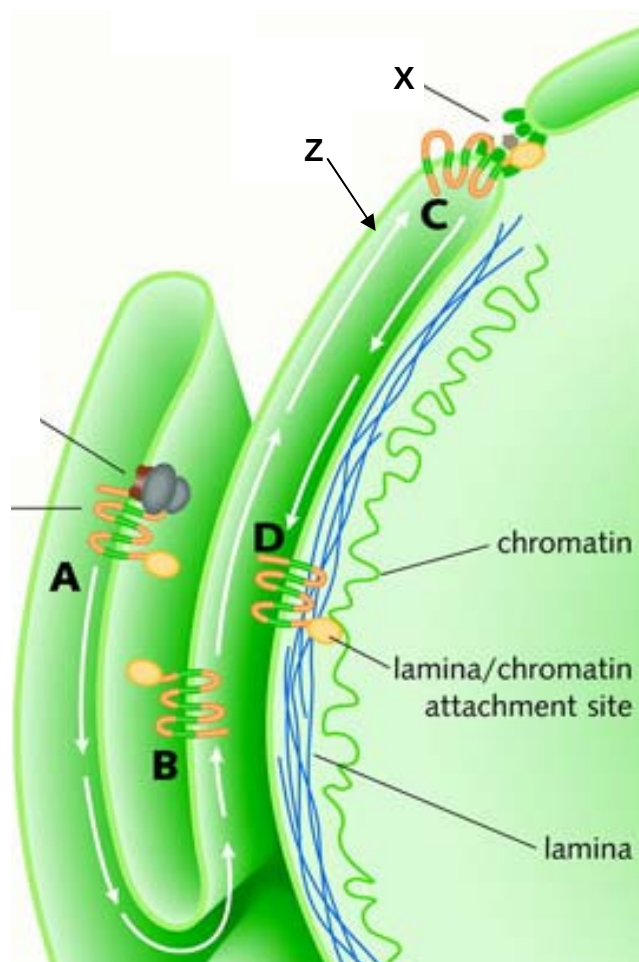


Fig. 1.2

(c) Briefly describe the role of X.

[1]

(d) Describe the structure in Region Z.

[2]

Fig. 1.3 shows a section of the cell membrane.

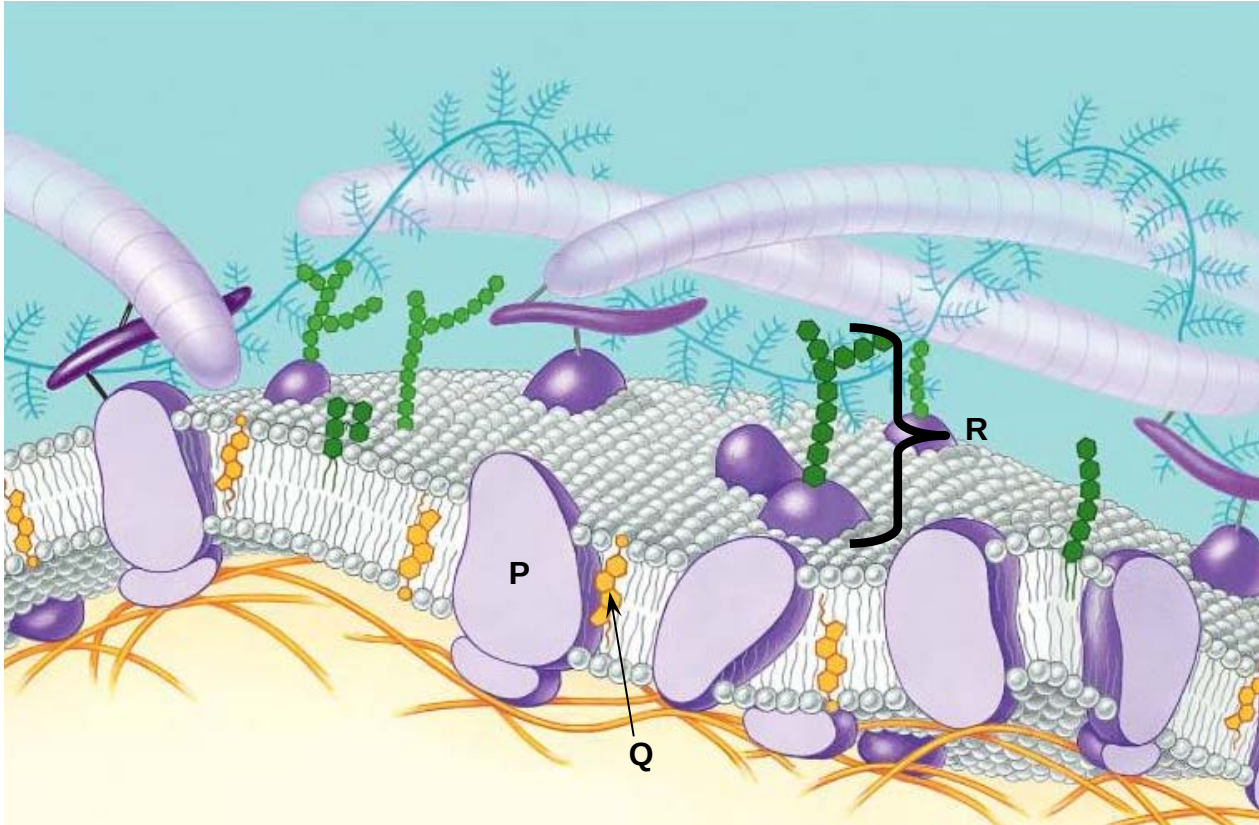


Fig. 1.3

(e) With reference to Fig. 1.3, identify the labelled structures and state their roles.

[3]

(f) State two possible functions of cell membrane.

[2]

[12 Marks]

2. Fig. 2.1 is a diagram showing the changes in membrane potential of a nerve cell.

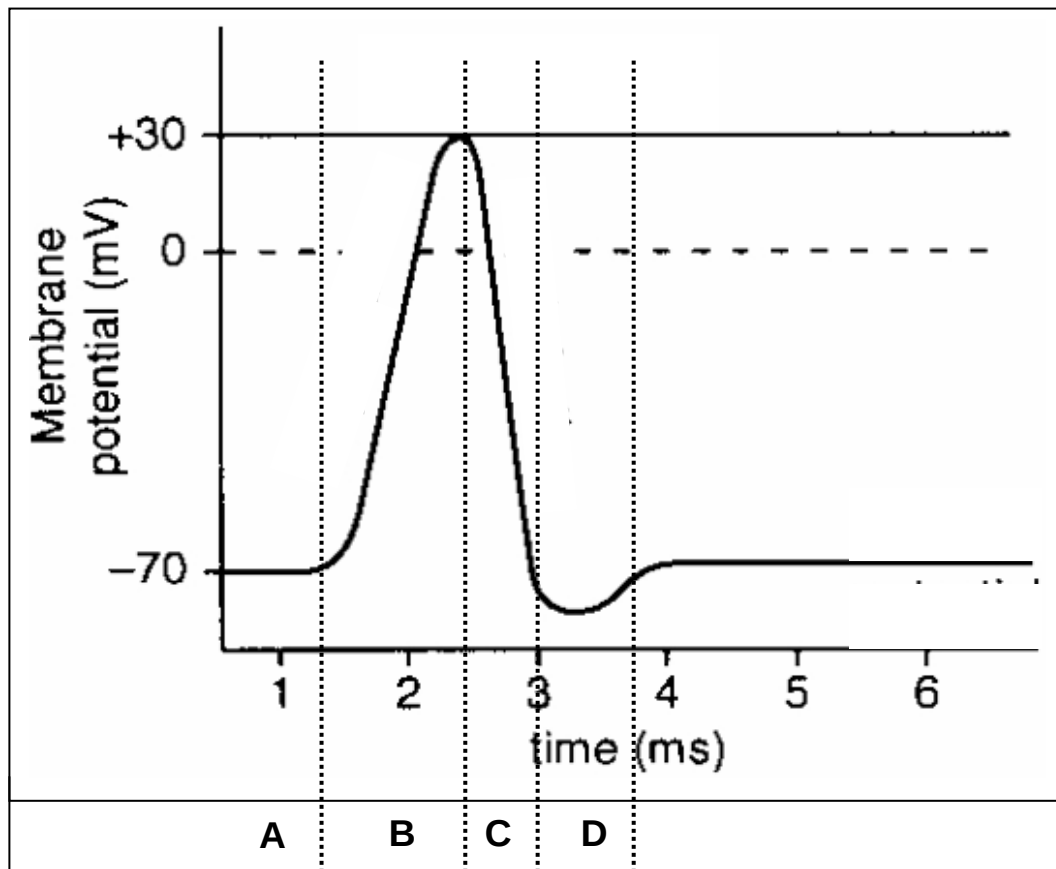


Fig. 2.1

- (a) Identify the stages **A** to **D** of an action potential.

[2]

A: _____

B: _____

C: _____

D: _____

(b) With reference to Fig. 2.1, explain Stages B and C.

[4]

Fig. 2.2 shows a molecular structure of tetrodotoxin, a potent neurotoxin with no known antidote. The binding site of this toxin is located at the pore opening of the voltage-gated Na^+ channel.

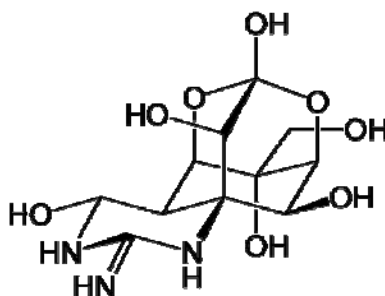


Fig. 2.2

(c) Suggest why tetrodotoxin is considered a lethal neurotoxin.

[3]

Some bacteria produce tetrodotoxin, which can be found in the internal organs of the pufferfish, as they share a symbiotic relationship. This makes them a possibly lethal meal for any predatory fish. Tetrodotoxin is found primarily in the ovaries and liver, although smaller amounts exist in the intestines and skin, as well as trace amounts in muscle tissue and blood.

The pufferfish has immunity to the poison due to a mutation in the protein sequence of the sodium-gated channel on the cell membranes.

(d) Suggest how this mutation confers immunity to tetrodotoxin in pufferfish. [1]

[10 Marks]

3.

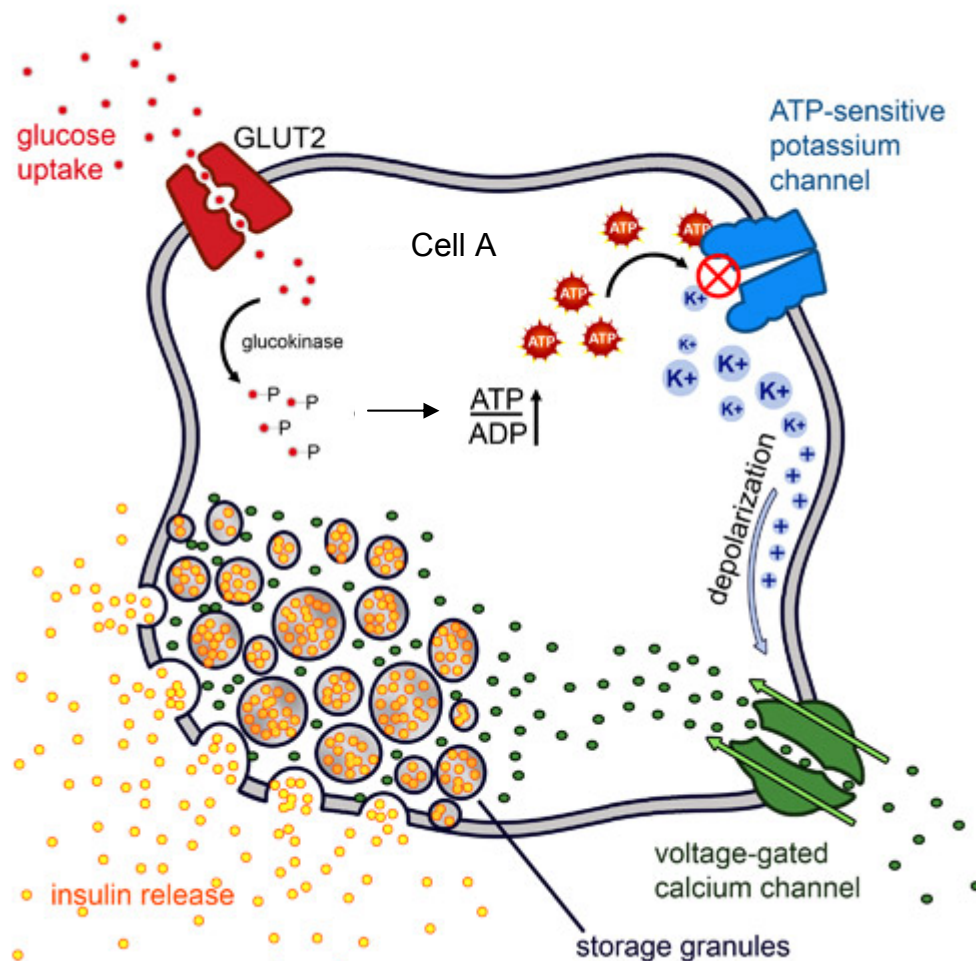


Fig. 3.1

Fig. 3.1 shows that insulin secretion in Cell A is triggered by rising blood glucose levels. The uptake of glucose causes a rise in the ATP:ADP ratio in the cell. This in turn closes the potassium channel and leads to depolarization of the membrane, causing the voltage-gated calcium channel to open. The influx of calcium ions leads to the exocytotic release of insulin from their storage granules.

(a) Identify Cell A. [1]

(b) Explain how GLUT2 allows the passage of glucose. [1]

- (c) Explain how dysfunctional GLUT2 in diabetic patients leads to their required regular injections of insulin. [3]

- (d) Suggest the reason for the increase of ATP:ADP ratio following the uptake of glucose through GLUT2. [2]

- (e) Explain why insulin is defined as an endocrine hormone. [1]

- (f) Outline the action of insulin in the signaling pathway upon reaching the liver and muscle cells. [4]

[12 Marks]

4. Fig. 4.1 shows a cell undergoing two different stages of meiosis (to the same scale).

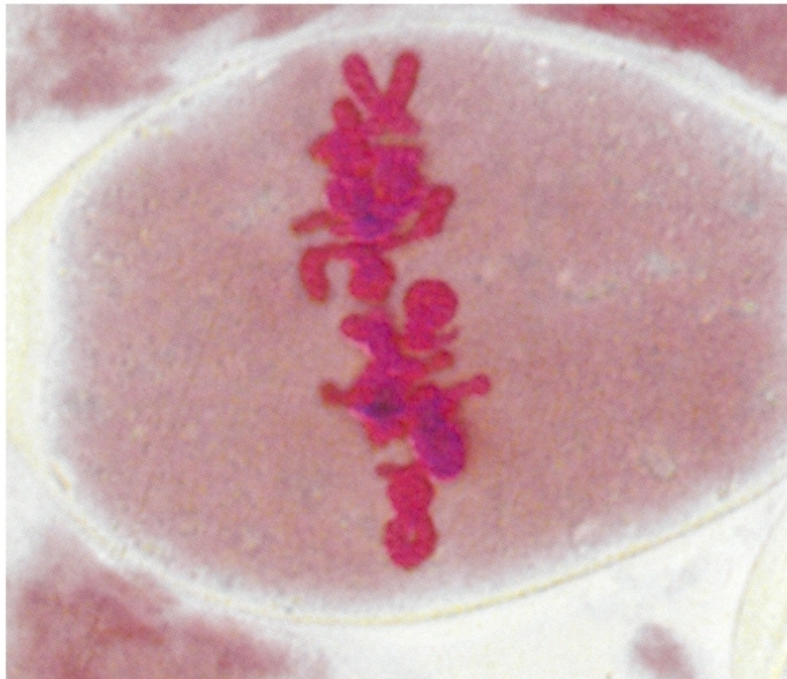
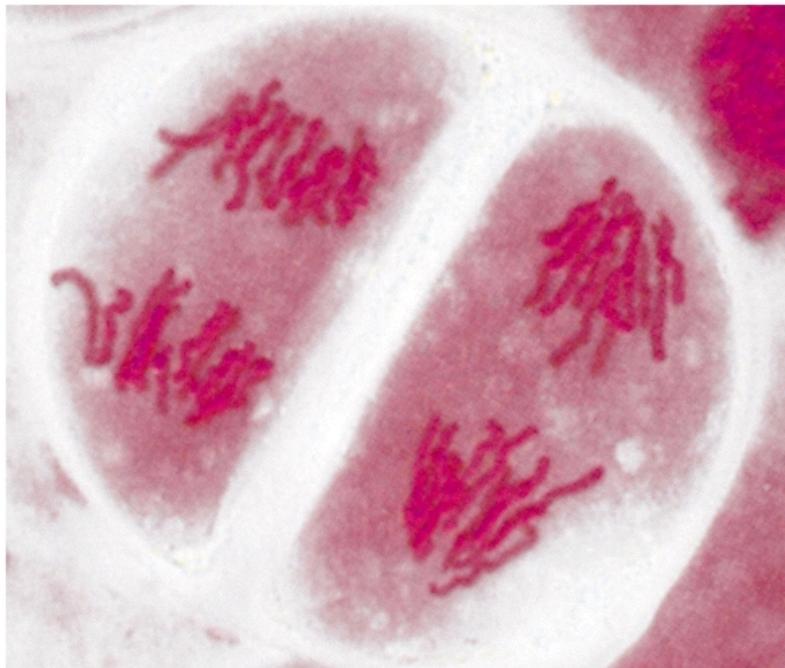
**A****B**

Fig. 4.1

(a) Identify stages **A** and **B**.

[2]

(b) Explain the phenomenon observed in **B**.

[2]

(c) Describe how meiosis can lead to genetic variation.

[2]

Okadaic acid is a toxin that causes spindle fibres to significantly increase in length and disrupts the interactions between microtubules and kinetochores, resulting in a disorganized metaphase plate.

(d) Suggest a possible outcome of a disorganised metaphase plate.

[2]

[8 Marks]

5. Hunter's syndrome is a serious genetic disorder and it usually affects males. It interferes with the body's ability to break down and recycle specific mucopolysaccharides, also known as glycosaminoglycans or GAG. The visible signs and symptoms of Hunter syndrome in younger people are usually the first clues leading to a diagnosis. In general, the time of diagnosis usually occurs from about 2 to 4 years of age.

Fig. 5.1 shows the pedigree of Mariah's family. Mariah (M) married Nick (N) and had three children. One of the children had Hunter's syndrome.

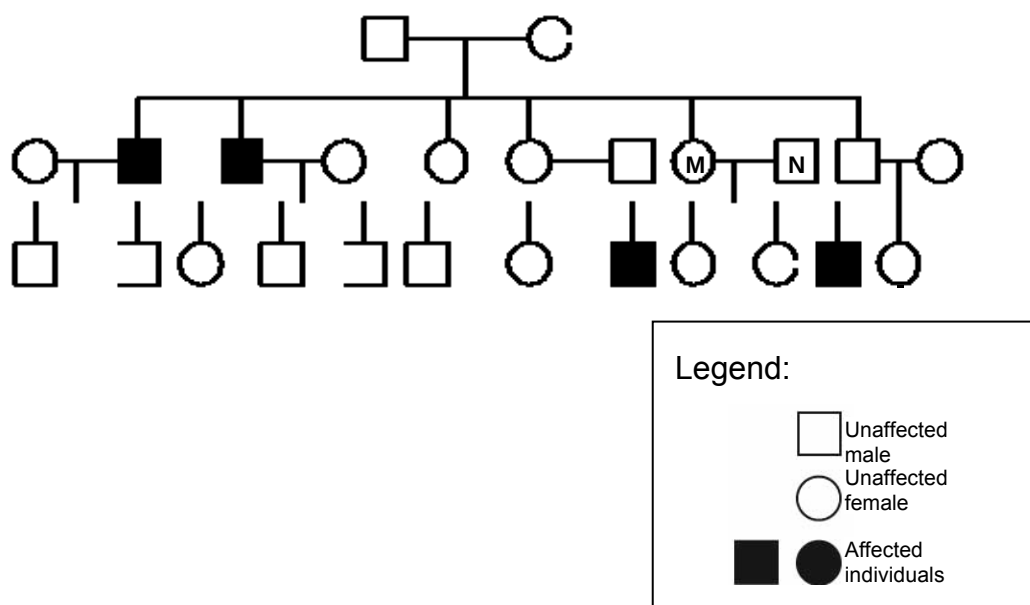


Fig. 5.1

- (a) With reference to Fig. 5.1, determine the type of inheritance for Hunter's syndrome. [1]

- (b) Using a genetic diagram, explain the type of inheritance pattern seen in the offspring of Mariah and Nick. [6]

A survey was conducted on the occurrence of Hunter's syndrome in Singapore. Information was gathered from 100 couples whereby the female is a carrier for Hunter's syndrome, while the male is normal. Only the first child of each couple was tested for Hunter's syndrome. In total, it was found that 6 children had the disease.

- (c) Use a suitable test with a 5% level of significance to determine whether the inheritance of the disease in the survey follows Mendelian's pattern of inheritance. [6]

(d) Suggest one possible reason why the survey conducted may not be a fair one.

[1]

[14 Marks]

6. In 1975 a team of Japanese scientists discovered a strain of *Flavobacterium* living in ponds containing waste water from a factory producing nylon. *Flavobacterium* are capable of digesting certain by-products of nylon, even though those substances are not known to have existed before the invention of nylon in 1935. This strain of *Flavobacterium* sp. K172, became popularly known as *nylon-eating bacteria*, and the enzymes are collectively known as *nylonase*. *Nylonase* were found to have no catalytic activity towards naturally occurring amide compounds, suggesting that the enzymes are completely new, as there is no homology in the amino acid sequence found between *nylonase* and other known enzymes.

Other species of *Flavobacterium* are known to commonly cause diseases in freshwater fishes as they derive their carbon and nitrogen sources from them. Due to the poisoning by factory wastes, the population of fish in the pond has declined.

- (a) It was suggested that the gene coding for *nylonase* may have arisen as a result of mutation. Suggest and explain the type of gene mutation that could have taken place. [2]

- (b) Account for the predominance of *nylon-eating bacteria* in the ponds through natural selection. [3]

- (c) It is unlikely for *nylon-eating bacteria* to be commonly found in freshwaters that are unpolluted. Explain why. [2]

- (d) Suggest how one can estimate quantitatively the phylogenetic distance between two evolutionarily related bacteria. [2]

- (e) Ecological speciation concept is generally preferred over biological speciation concept in classifying bacteria such as *Flavobacterium*. Explain why. [1]

- (f) The classification of *Flavobacterium gondwanense* is listed as shown below in decreasing levels of taxa. Fill in the blanks. [2]

Levels of taxa	Name of Taxa
Kingdom	Bacteria
	Bacteriodetes
Class	Flavobacteria
	Flavobacteriaceae
Family	Flavobacteriaceae
Genus	
Species	

[12 Marks]

7. The *lac* operon consists of three structural genes, a promoter, a terminator, a regulator, and an operator as shown in Fig. 7.1. The three structural genes are: *lacZ*, *lacY*, and *lacA*. *LacZ* encodes β -galactosidase, an intracellular enzyme that cleaves the disaccharide lactose into glucose and galactose. *LacY* encodes β -galactoside permease, a membrane-bound transport protein that pumps lactose into the cell. *LacA* encodes β -galactoside transacetylase, an enzyme that transfers an acetyl group from acetyl-CoA to β -galactosides.

Only *lacZ* and *lacY* appear to be necessary for lactose catabolism. The *lacI* shown in Fig. 7.1 is sometimes also considered as part of the *lac* operon.

Bacteria usually prefer to use glucose over lactose as source of energy.



Fig. 7.1

- (a) Explain why bacteria prefer glucose over lactose as energy source. [2]

- (b) Describe how the presence of allolactose turns the expression of the *lac* operon on. [2]

(c) State the function of the following in the regulation of *lac* operon expression.

(i) Promoter

[1]

(ii) *LacI*

[1]

Fig. 7.2 shows the polycistronic nature of *lac* operon.

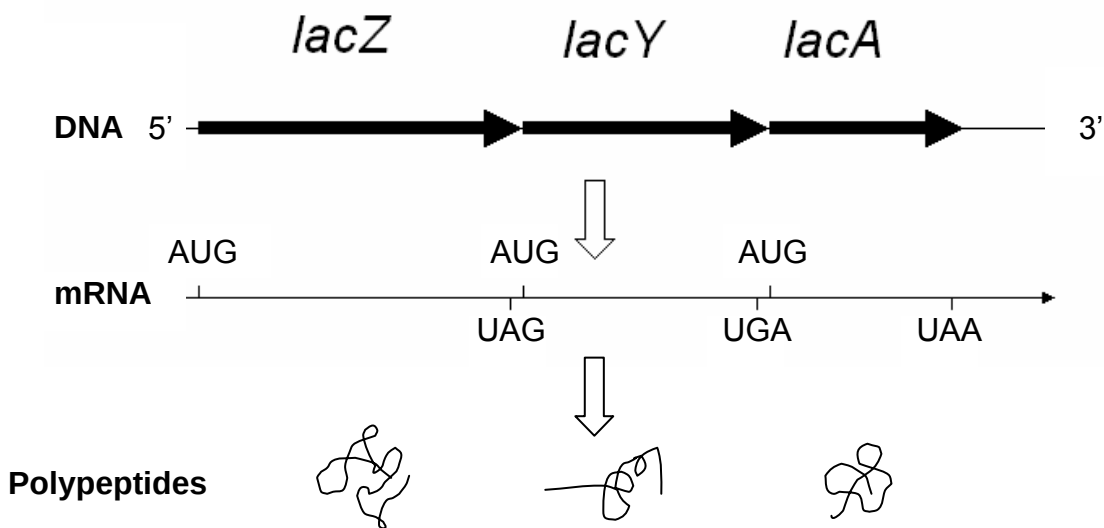


Fig. 7.2

(d) Explain how Fig. 7.2 illustrates the polycistronic nature of the *lac* operon.

[2]

A certain bacteriophage is known to insert its own genome between the *lacZ* and *lacY* structural genes to give prophages.

(e) Explain the change in number of bacteriophage (if any) in conditions when host bacteria are grown in environment with

(i) abundant supply of glucose and no lactose supply. [2]

(ii) no supply of glucose and abundant supply of lactose. [2]

[12 Marks]

Section B

Answer **one** question.

Write your answers on a separate answer paper provided.

Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answers should be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated by the question.

- 8 (a) Outline the main stages of Calvin Cycle. [6]
- (b) Explain the importance of oxidative phosphorylation. [8]
- (c) Explain what happens in an animal cell when there is insufficient oxygen for aerobic respiration. [6]
- 9 (a) Describe the process of transcription in prokaryotes. [6]
- (b) Explain how the amount of proteins in a eukaryotic cell can be controlled. [8]
- (c) Describe and explain the features essential in maintaining the structure of DNA double helix. [6]

Annex

Formula Sheet

Standard deviation

$$s = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}}$$

χ^2 test

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

$v = c - 1$

t-test

$$t = \frac{|\bar{X}_A - \bar{X}_B|}{\sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}}}$$

$v = n_A + n_B - 2$

standard error

$$S_M = \frac{s}{\sqrt{n}}$$

- s = standard deviation
- Σ = sum of
- X_i = individual value
- $\bar{X}$ = mean of the set of values
- n = sample size
- v = degrees of freedom
- c = number of classes
- O = observed 'value'
- E = expected 'value'

Table 5: Critical Values for the Chi-Square (χ^2) Test

Level of significance (P)					
	0.05	0.025	0.01	0.005	0.001
1	3.84	5.02	6.63	7.88	10.83
2	5.99	7.38	9.21	10.60	13.81
3	7.81	9.35	11.34	12.84	16.27
4	9.49	11.14	13.28	14.86	18.47
5	11.07	12.83	15.09	16.75	20.52
6	12.59	14.45	16.81	18.55	22.46
7	14.07	16.01	18.48	20.28	24.32
8	15.51	17.53	20.09	21.96	26.13
9	16.92	19.02	21.67	23.59	27.88
10	18.31	20.48	23.21	25.19	29.59
11	19.68	21.92	24.73	26.76	31.26
12	21.03	23.34	26.22	28.30	32.91
13	22.36	24.74	27.69	29.82	34.53
14	23.68	26.12	29.14	31.32	36.12
15	25.00	27.49	30.58	32.80	37.70
16	26.30	28.85	32.00	34.27	39.25
17	27.59	30.19	33.41	35.72	40.79
18	28.87	31.53	34.81	37.16	42.31
19	30.14	32.85	36.19	38.58	43.82
20	31.41	34.17	37.57	40.00	45.32
21	32.67	35.48	38.93	41.40	46.80
22	33.92	36.78	40.29	42.80	48.27
23	35.17	38.08	41.64	44.18	49.73
24	36.42	39.36	42.98	45.56	51.18
25	37.65	40.65	44.31	46.93	52.62
26	38.89	41.92	45.64	48.29	54.05
27	40.11	43.19	46.96	49.64	55.48
28	41.34	44.46	48.28	50.99	56.89
29	42.56	45.72	49.59	52.34	58.30
30	43.77	46.98	50.89	53.67	59.70
40	55.76	59.34	63.69	66.77	73.40
50	67.50	71.42	76.16	79.49	86.66
60	79.08	83.30	88.38	91.95	99.61
70	90.53	95.02	100.43	104.22	112.32
80	101.88	106.63	112.33	116.32	124.84
90	113.15	118.14	124.12	128.30	137.21
100	124.34	129.56	135.81	140.17	149.44

Table 1: Critical Values for the t-Test

	Level of significance (P)					
	0.05	0.025	0.01	0.005	0.001	0.0005
Degrees of freedom (df)						
1	6.314	12.706	31.821	63.657	318.31	636.62
2	2.920	4.303	6.965	9.925	22.327	31.598
3	2.353	3.182	4.541	5.841	10.214	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.869	3.385	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
21	1.721	2.080	2.518	2.831	3.527	3.819
22	1.717	2.074	2.508	2.819	3.505	3.782
23	1.714	2.069	2.500	2.807	3.485	3.767
24	1.711	2.064	2.492	2.797	3.467	3.745
25	1.708	2.060	2.485	2.787	3.450	3.725
26	1.706	2.056	2.479	2.779	3.435	3.707
27	1.703	2.052	2.473	2.771	3.421	3.690
28	1.701	2.048	2.467	2.763	3.408	3.674
29	1.699	2.045	2.462	2.756	3.396	3.659
30	1.697	2.042	2.457	2.750	3.385	3.646

Table 1 (contd): Critical Values for the t-Test

	Level of significance (P)					
	0.05	0.025	0.01	0.005	0.001	0.0005
Degrees of freedom (df)						
31	1.696	2.040	2.453	2.744	3.375	3.634
32	1.694	2.037	2.449	2.739	3.366	3.628
33	1.693	2.034	2.445	2.733	3.356	3.611
34	1.692	2.032	2.441	2.728	3.348	3.601
35	1.689	2.030	2.437	2.723	3.340	3.591
36	1.688	2.028	2.434	2.719	3.333	3.582
37	1.687	2.026	2.431	2.715	3.326	3.574
38	1.686	2.024	2.429	2.712	3.319	3.566
39	1.685	2.022	2.426	2.708	3.313	3.558
40	1.684	2.021	2.423	2.704	3.307	3.551
41	1.683	2.020	2.421	2.701	3.301	3.544
42	1.682	2.018	2.418	2.698	3.296	3.538
43	1.681	2.017	2.416	2.695	3.291	3.532
44	1.680	2.016	2.414	2.692	3.286	3.526
45	1.680	2.014	2.412	2.690	3.281	3.520
46	1.679	2.013	2.140	2.687	3.277	3.515
47	1.678	2.012	2.408	2.685	3.273	3.510
48	1.677	2.011	2.406	2.682	3.269	3.505
49	1.677	2.010	2.405	2.680	3.265	3.500
50	1.676	2.009	2.403	2.678	3.261	3.496
51	1.675	2.008	2.401	2.676	3.258	3.492
52	1.675	2.007	2.400	2.674	3.254	3.488
53	1.674	2.006	2.399	2.672	3.251	3.484
54	1.674	2.005	2.397	2.670	3.248	3.480
55	1.673	2.004	2.396	2.668	3.245	3.476
56	1.673	2.003	2.395	2.666	3.242	3.473
57	1.672	2.002	2.393	2.665	3.240	3.469
58	1.672	2.002	2.392	2.663	3.237	3.466
59	1.671	2.001	2.391	2.662	3.234	3.463
60	1.671	2.000	2.390	2.660	3.232	3.460

One-tailed test
Two-tailed test